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Lab	03
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LAB MANUAL: CABLE TESTING USING RJ45/RJ11 AND BNC CABLE TESTER

OBJECTIVE

To learn how to test network cables (RJ45, RJ11) and BNC cables for continuity, proper wiring, and faults using a cable tester.

APPARATUS / MATERIALS REQUIRED

- Network cable tester (RJ45/RJ11/BNC compatible)
- RJ45 connectors (8P8C)
- RJ11 connectors (6P4C)
- BNC connectors (for coaxial cable)
- CAT5e / CAT6 twisted pair cables
- Coaxial cable (for BNC testing)
- Crimping tool for RJ45/RJ11/BNC
- Cable stripper / cutter
- Lab sheet, pen, glue/tape

THEORY

A cable tester verifies the integrity of cables by checking:

- Connected: All pins are properly wired.
- No Connection: Open circuit or broken wires.
- Non-parallel / Miswire: Pins connected incorrectly.
- Short: Two or more wires touching incorrectly.
- Impedance Check (BNC): Verifies 50Ω/75Ω termination.
- Battery Status: Green = Good, Yellow = Low.

The tester applies a signal through the cable and indicates results via LEDs.

PROCEDURE

A. Testing RJ45 / RJ11 Cables

1. Power on the cable tester (Green LED = Battery OK).
2. Connect both ends of the cable to main and remote units.
3. Observe LED indicators (Connected, No Connection, Non-Parallel, Short).
4. Record the result.

B. Making a Straight-Through RJ45 Cable

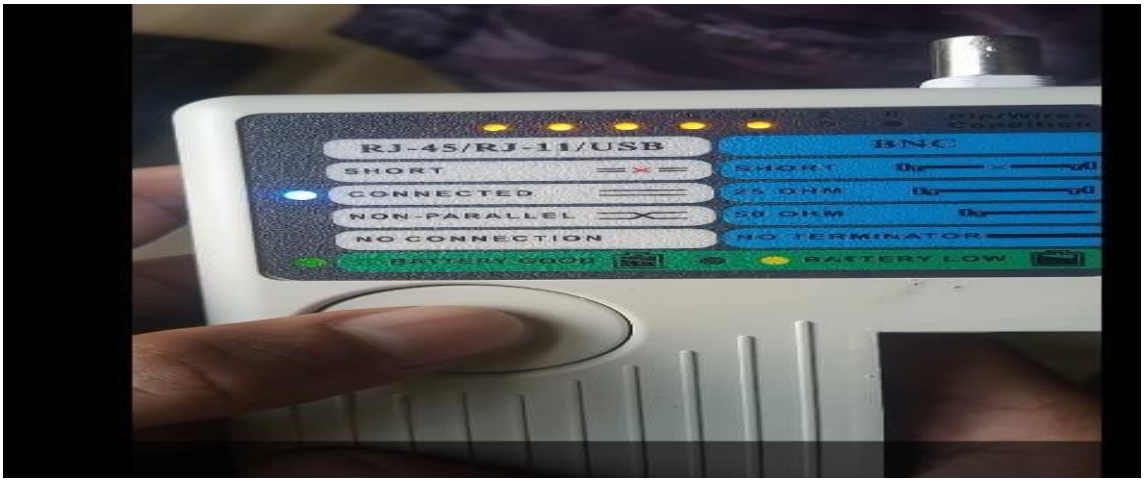
Follow either wiring standard:

- T568A: White/Green, Green, White/Orange, Blue, White/Blue, Orange, White/Brown, Brown
- T568B: White/Orange, Orange, White/Green, Blue, White/Blue, Green, White/Brown, Brown

5. Strip cable jacket (~1 inch).
6. Arrange wires in correct order.
7. Trim wires evenly.
8. Insert into RJ45 connector.
9. Crimp using tool.
10. Test using cable tester.

C. Testing BNC Coaxial Cable

11. Connect cable to tester.
12. Attach terminator or remote.
13. Observe LED indicators (50Ω, 75Ω, Short, No Terminator).
14. Record results.



PRECAUTIONS

- Ensure tester battery is charged.
- Avoid bending cables excessively.
- Follow correct wiring standards.
- Check LED indicators carefully.
- Ensure connectors are properly crimped.

CONCLUSION

Cables showing CONNECTED are suitable for use.

Faulty cables (Short, Open, Miswire) must be re-terminated.

Cable testers ensure reliable network connections.